DO NOW — RECALL

Specification gap and misconception check

Name: _____ Class: _____ Date: _____

1. What does a salt add to password storage?

2. What is an INNER JOIN result based on?

3. Which OS mechanism translates pages to frames?

4. Full adder A=1,B=0,carry-in=0: sum and carry, and binary total.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A distinct random per-account value, improving resilience of password hashing.

2. Model answer

Matching records according to a join condition.

3. Model answer

Page table.

4. Model answer

$1+0+0=1=01_2$; sum 1, carry 0.

Vocabulary — know these terms

misconception — Incorrect mental model of a mechanism.

counterexample — Case disproving a general claim.

transfer — Apply understanding to new case.

diagnostic — Detect specific gaps in knowledge.

NEW LEARNING

Read, annotate and explain

Understand

When correcting an error, identify the incorrect assumption precisely; a bare True/False is weak diagnostic evidence.

Trace

Connect each correction to a worked example: HTTPS versus phishing, paging versus remote files, key joins versus unmatched rows.

Evaluate

Show transfer by applying the rule to a different problem without being prompted by the original misconception.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Correct 'a page fault is a missing website page'.

Predict first

A page fault is a memory-management event where required virtual page data is not currently resident.

Show method

HTTP 404 is a web server status about a requested resource; the word 'page' refers to different concepts.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Correct four statements: 'DNS sends HTML'; 'ROM is volatile'; 'SHA is encryption'; 'INNER JOIN keeps all rows'. [8]

Question 2

Explain why a school's cloud files and its RAM paging system should not be described by one mechanism. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Choose three of the previous misconceptions and construct a counterexample for each. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An accurate definition alone guarantees that you can apply a concept.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.